

Wang Lei

Gender: Male Objective: New Graduate Backend Engineer Location: Beijing

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Education

2021.09 - 2025.06 Central South University Electronic Information Engineering (B.S.)

Courses: Operating Systems, Design Patterns, Database Systems, Machine Learning, Data Structures and Algorithms, Distributed Systems

GPA 3.3/4.0, top 10% in major; merit scholarship

Summary

New CS graduate with two backend internships, solid Java/Spring Boot and database fundamentals and full project experience.

Internship Experience

2024.06 - 2024.12 Trip.com Intern Backend Engineer

- Owned backend of core transaction and order services, splitting the monolith into order, settlement and risk-control microservices.
- Built service observability covering API latency, error rate, JVM GC, thread pools and consumer backlog with Prometheus and Grafana, significantly lowering maintenance cost.
- Tuned Redis caching for hot keys and cache breakdown with local cache and distributed locks, raising hit rate above 90%, improving end-to-end delivery quality.

2023.07 - 2023.09 ByteDance Intern Backend Engineer

- Led a production incident review with root-cause analysis and drove an alert-tiering and on-call rotation mechanism, ensuring online service stability.
- Rebuilt the config and template management service to support hot template reload without restart, cutting release time by 43%.
- Investigated rising instance memory RSS using GC logs, heap dump and arthas, found an unbounded cache and cut memory by 52%, improving end-to-end delivery quality.

Projects

Distributed Payment & Settlement Platform (Spring Boot + MySQL + Kafka + Redis)

- Applied idempotency and eventual consistency to avoid duplicate charges on retry, and related standards were adopted by the team.
- Used sharding and read-write splitting to keep single-table rows within tens of millions.
- Recorded an audit trace ID for every payment state transition for compliance, improving end-to-end delivery quality.
- Implemented daily reconciliation across payment gateway, ledger and settlement files with auto-correction, with experience documented and shared internally.

High-Concurrency Flash-Sale & Inventory System (Java + Redis + RocketMQ)

- Stayed stable at 20000 QPS in load tests with error rate below 0.1%.
- Built circuit breaking and fallback for downstream failures, recognized by stakeholders and peers.

- Pre-deducted inventory in Redis with async DB writes to absorb traffic spikes.
 - Designed distributed locks and token-bucket rate limiting to prevent oversell.
- Enterprise Auth & Gateway Platform (Spring Cloud Gateway + OAuth2)
- Shipped a unified SDK to lower integration cost across teams.
 - Added circuit breaking to isolate faulty downstreams.
 - Integrated OAuth2 and RBAC with multi-tenant isolation.
 - Unified auth, rate limiting, gray routing and tracing as reusable platform capabilities, and key metrics improved steadily.

Highlights

- Established metrics and reviews around design patterns and refactoring to keep improving delivery quality
- Built a reusable methodology and docs around MySQL indexing and slow-query tuning, adopted by several teams
- Continuously optimized Distributed transactions and consistency, cutting key-path latency from 900ms to 380ms
- Owned core modules that steadily served 6220K daily visits with 99.9%+ availability
- Drove a focused effort on computer networking basics, reducing related issues by about 43%

Skills

Core skills: Redis, Data Structures, JVM tuning and troubleshooting, Spring Boot

Proficient in: MySQL indexing and slow-query tuning, Kafka/RocketMQ messaging, MySQL, Docker and containerization

Familiar with: Distributed transactions and consistency, Java, Git, Redis caching and distributed locks, Computer Networks

Self Evaluation

- Enjoys retrospectives, cares about code quality and observability, and makes data-driven decisions
- Curious about new tech, continuously learning LLM and cloud-native topics and applying them in practice